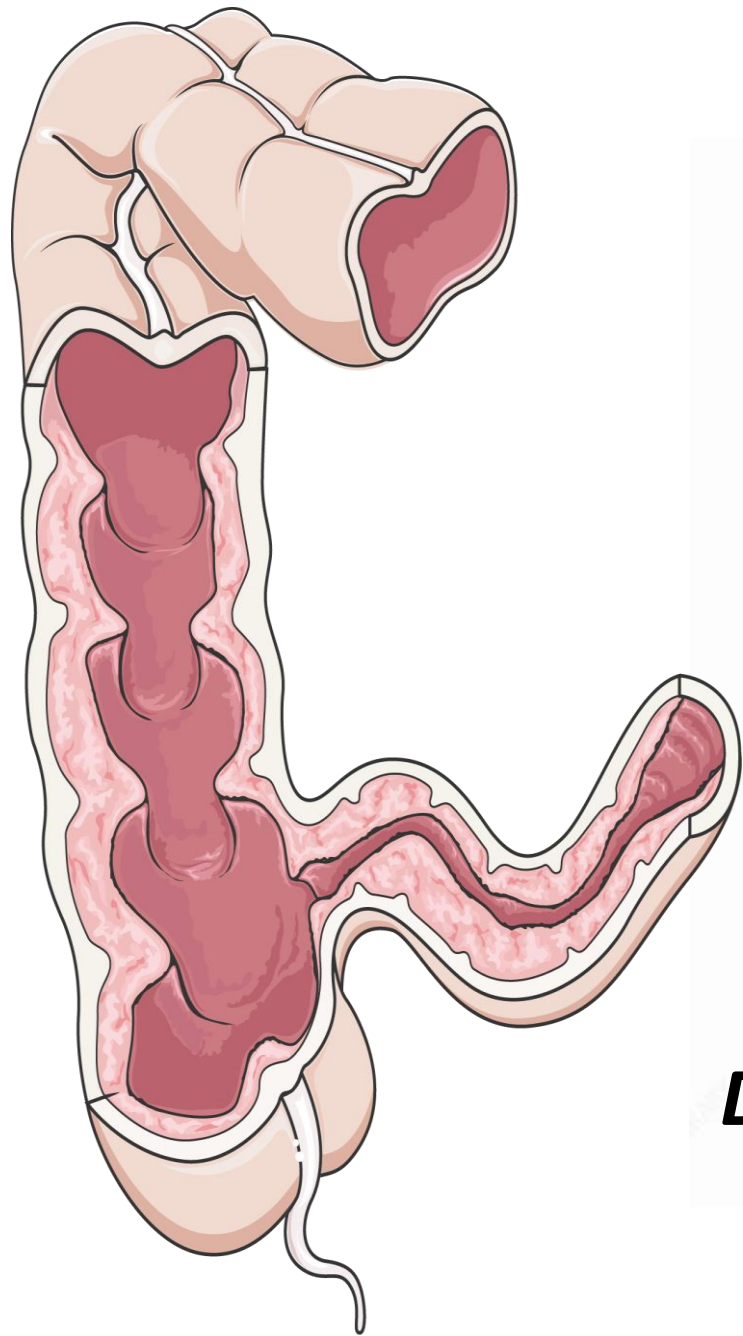
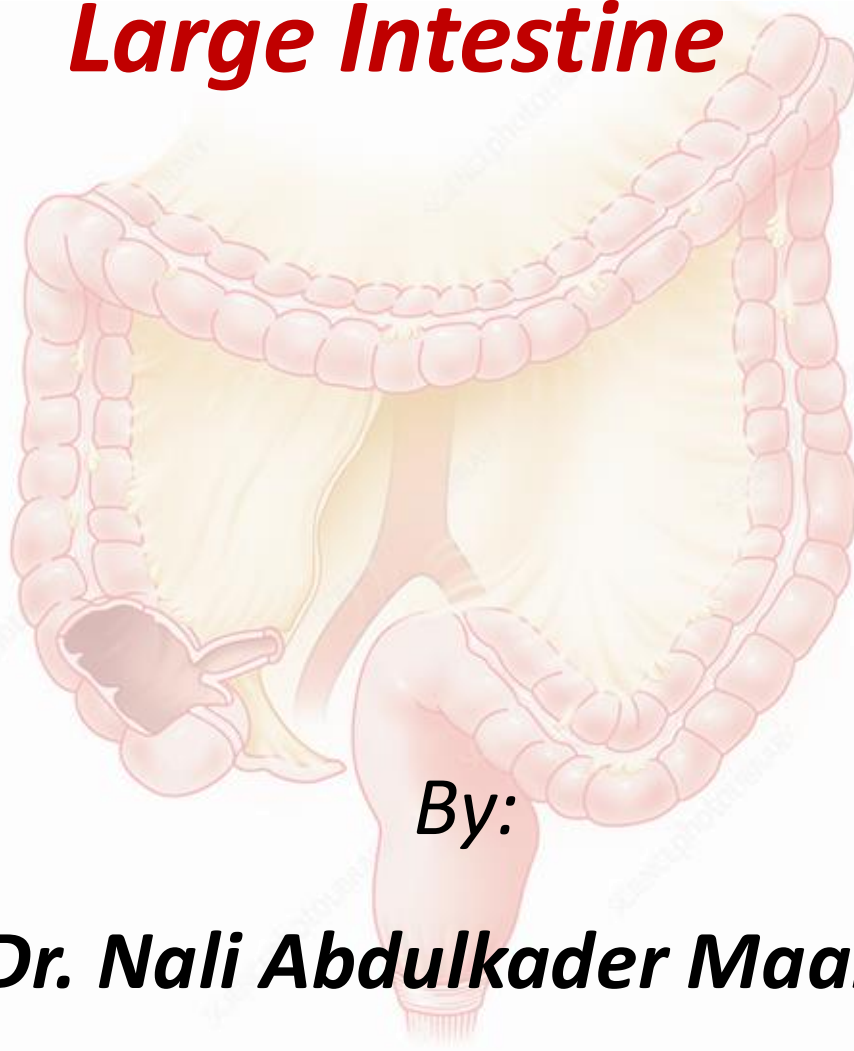


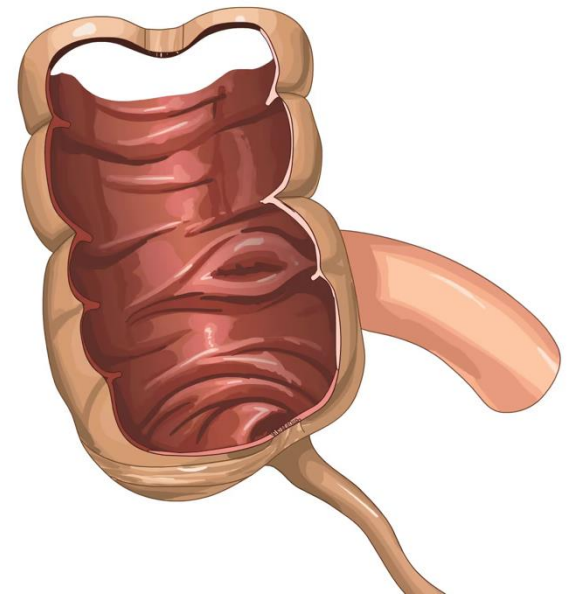
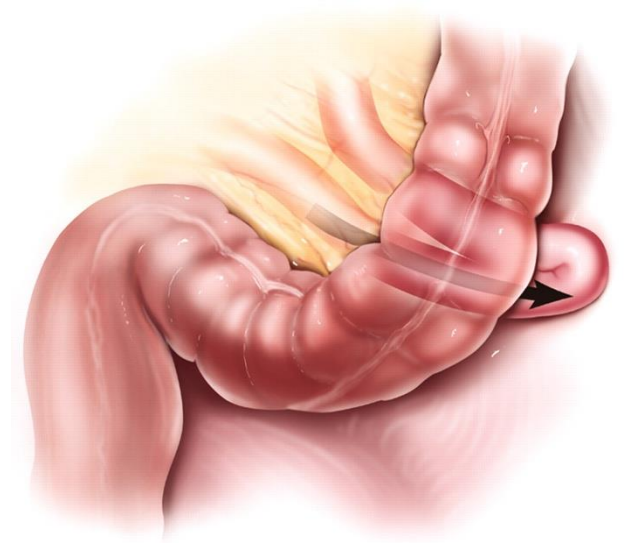


Large Intestine



By:

Dr. Nali Abdulkader Maaruf



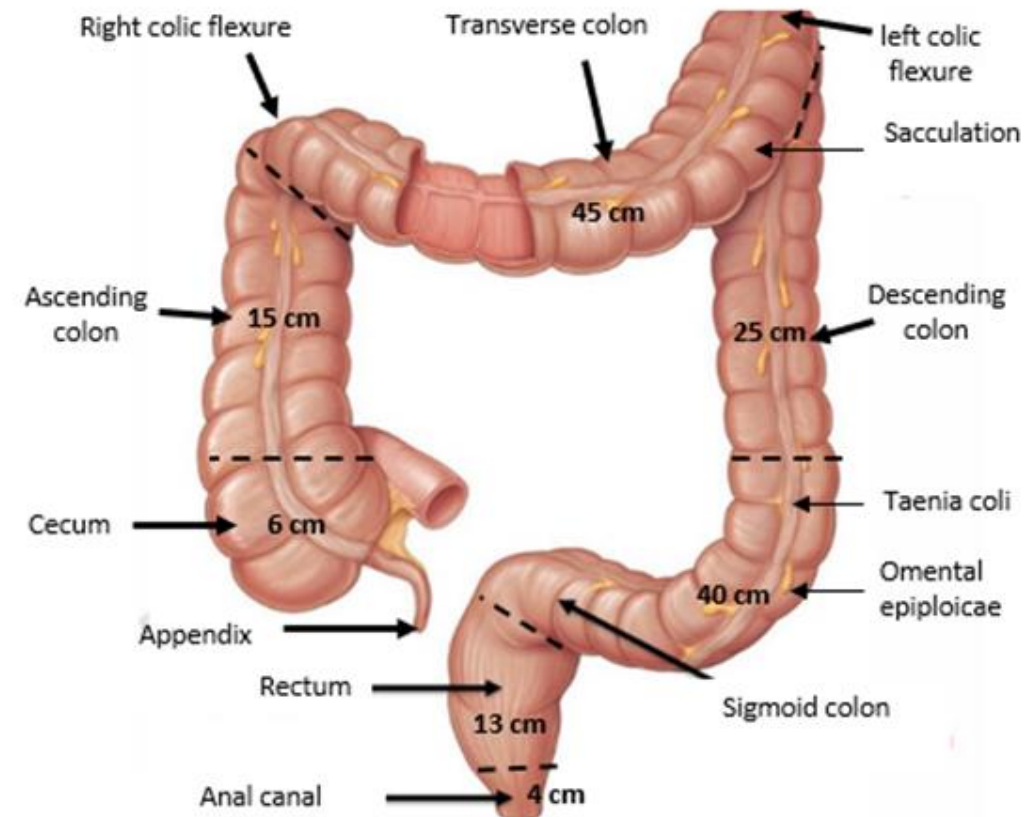
Objectives

At the end of the lecture, students should:

- ❖ List the parts of the large intestine.
- ❖ Describe the peritoneal covering and relations of each part of the large intestine (cecum, appendix, ascending, transverse, descending, sigmoid colon, rectum, and anal canal).
- ❖ Describe the arterial supply, venous drainage, lymphatic drainage, and nerve supply of the large intestine.
 - List the organs that can be examined during a digital rectal examination.
 - Differentiate between the upper and lower parts of the anal canal.
 - Describe anal sphincters.
 - Classify the types of hemorrhoids

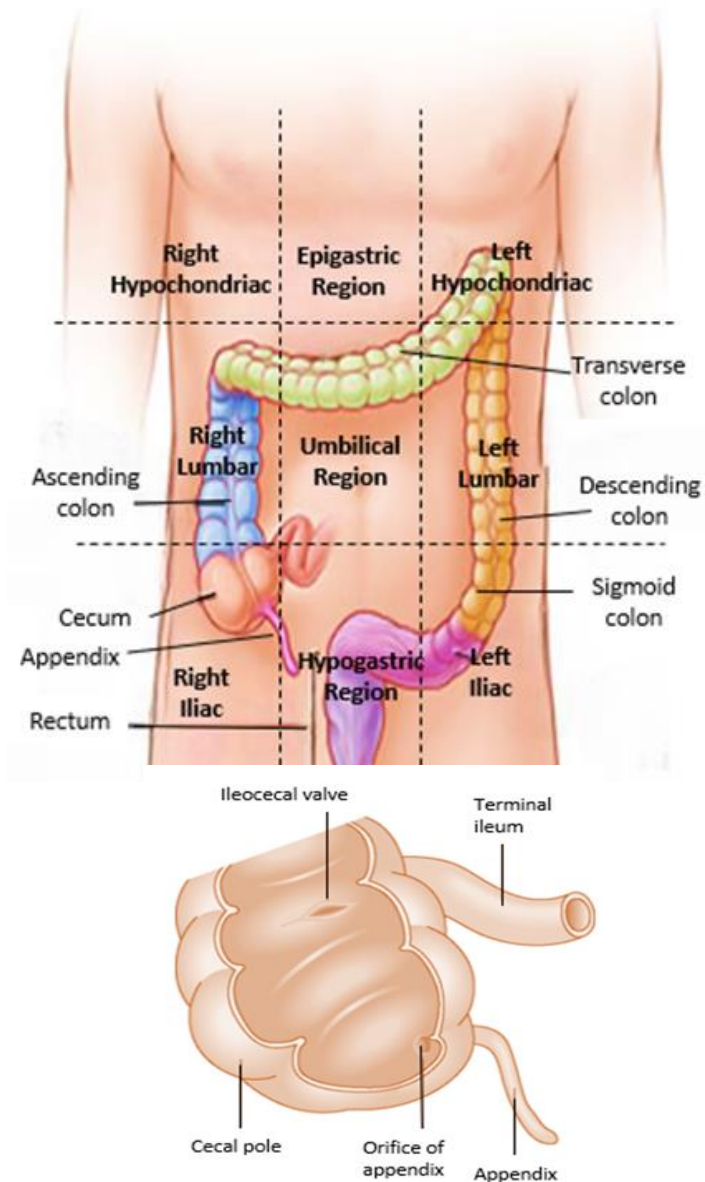
The large intestine

- ❑ The large intestine is about 1.5 m long and extends from the cecum in the right iliac fossa to the anus in the perineum.
- ❑ For descriptive purposes, the large intestine is divided into the following parts:
 - ❑ Cecum and appendix, colon, rectum, and anal canal
 - ❑ The colon is further divided into four parts: ascending, transverse, descending, and sigmoid colon.
 - ❑ All parts of the large intestine are retroperitoneal and fixed except the Vermiform appendix, transverse colon, and sigmoid colon, which are intraperitoneal and possess mesenteries.



CECUM

- ❑ The caecum (L. caecum = blind) is a dilated blind sac located in the right iliac fossa.
- ❑ Dimensions: Length: 6 cm (2½ inches); Width: 7.5 cm (3 inches), with the width being greater than the length, and represents the area of highest pressure of the colon.
- ❑ The posteromedial aspect is the opening of the ileum and is guarded by a valve known as the ileocecal valve as well as a sphincter referred to as the ileocecal sphincter. These structures regulate the flow of contents from the ileum to the caecum and prevent the regurgitation of caecal contents back into the ileum.

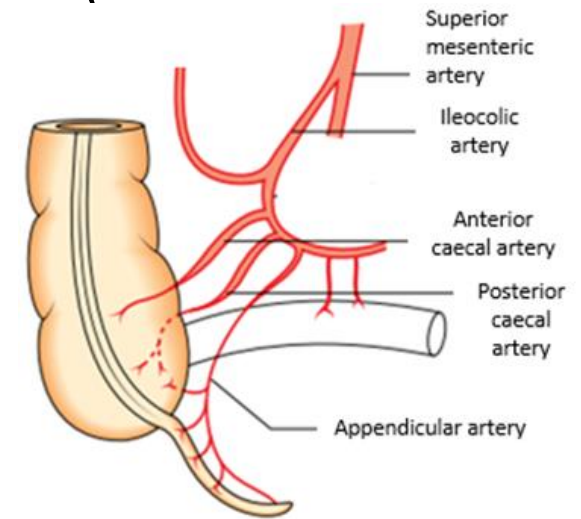


❖ Relations

- ❑ **Anterior:** coils of the small intestine, greater omentum, and anterior abdominal wall
- ❑ **Posterior:** Right psoas major and iliacus muscles, femoral nerve, lateral cutaneous nerve of the thigh, genitofemoral nerve of the right side, right gonadal vessels, and retrocecal recess (often contains the vermiform appendix)

❖ Blood Supply and Lymphatic Drainage

- ❑ The caecum receives its arterial blood supply from the ileocolic artery, which is a branch of the superior mesenteric artery
- ❑ The veins drain into the superior mesenteric vein, which subsequently empties into the portal system.
- ❑ Lymphatic vessels from the caecum drain into the superior mesenteric group of pre-aortic lymph nodes.

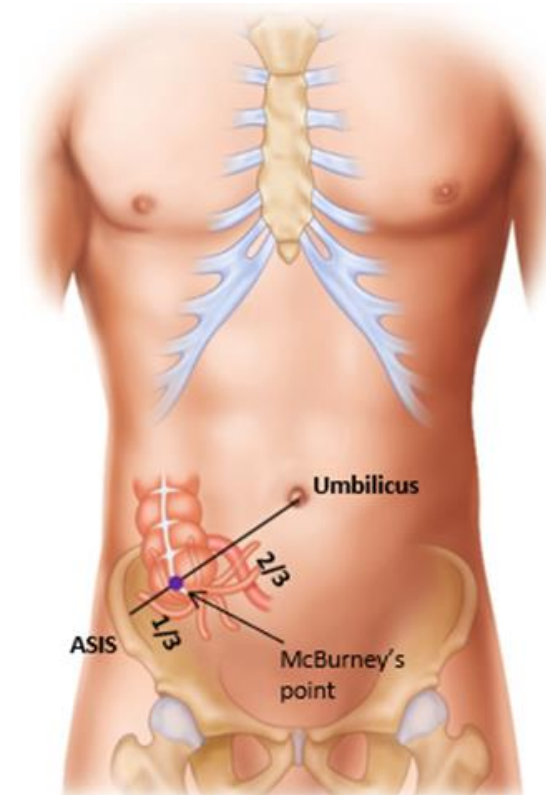


CLINICAL ANATOMY

- The caecum serves as a reference point for surgeons during operations for intestinal obstruction. If the caecum is distended, the obstruction is in the large intestine; if empty, it's in the small intestine

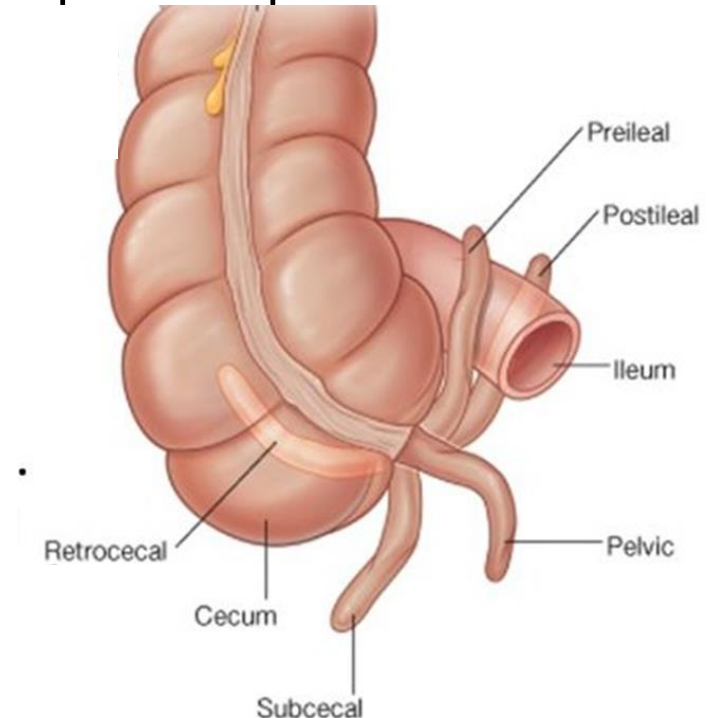
APPENDIX:

- ❑ The appendix is a narrow worm-like diverticulum, which arises from the posteromedial wall of the caecum about 2 cm below the ileocecal junction.
- ❑ It varies in length from 2 to 20 cm (average 9 cm).
- ❑ The appendix presents three parts—base, body, and tip.
- ❑ The base is the fixed part and is attached to the posteromedial wall of the caecum (McBurney's point). All three teniae of the caecum converge at the base of the appendix. This anatomical fact serves as a guide to the surgeon to search for the appendix during appendicectomy.
- ❑ The body is a narrow tubular part between the base and the tip.
- ❑ The tip is the least vascular distal blind end; it is variable in position



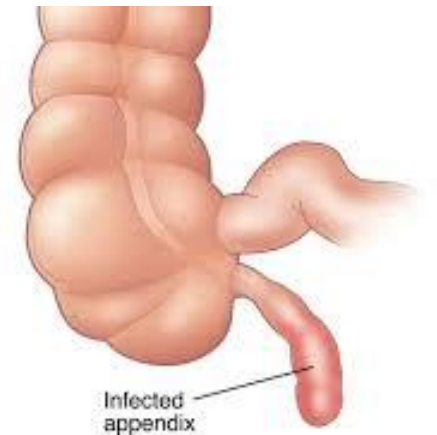
Common position of the tip of the appendix

- a) **Paracolic position:** in 2% of the cases.
- b) **Retrocecal/retrocolic position:** It is the commonest position (65.28%) of the appendix.
- c) **Pre-ileal) or behind (post-ileal) :** The appendix passes upward and medially in front of the terminal part of the ileum. The tip of the appendix points toward the spleen. The pre-ileal position is the most dangerous because inflammation from the appendix spreads into the general peritoneal cavity. The pre-ileal position occurs in 1% of the cases and post-ileal in 0.4% of the cases.
- d) **Pelvic position:** It is the second commonest position (31.01%).
- e) **Subcaecal position:** It occurs in 2% of the cases.



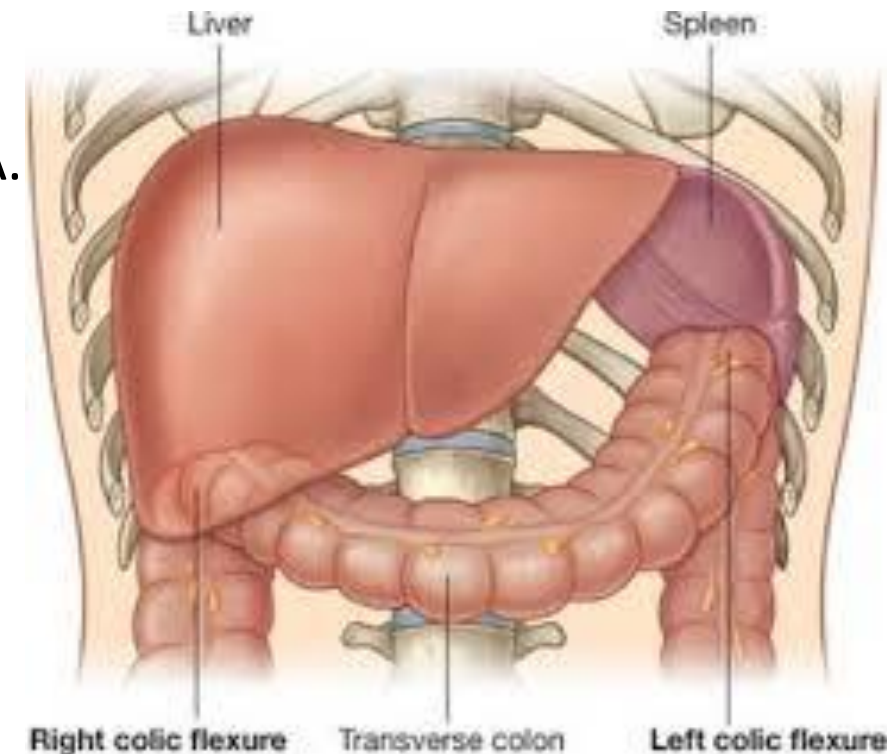
CLINICAL ANATOMY

- The inflammation of the appendix is known as appendicitis, and acute appendicitis represents a common surgical emergency.
- Initially, the pain associated with appendicitis is called referred pain, often felt in the umbilical region. This occurs because both areas share the same segmental nerve supply (specifically, the T10 spinal segment) due to visceral pain irritation.
- Upon palpation, maximum tenderness is typically elicited at McBurney's point.



ASCENDING COLON

- ❑ The ascending colon is a continuation of the cecum.
- ❑ It is approximately 6 inches (15 cm) long and extends from the cecum, to the inferior surface of the right lobe of the liver, where it bends to the left to form the hepatic (right colic) flexure.
- ❑ It is located in the right paracolic gutter
- ❑ **Blood supply** from the ileocolic and right colic branches of the SMA.
- ❑ **The veins** follow the same course as the arteries and drain into the superior mesenteric vein.
- ❑ **Lymphatic drainage** is via lymph vessels that drain into the superior mesenteric nodes

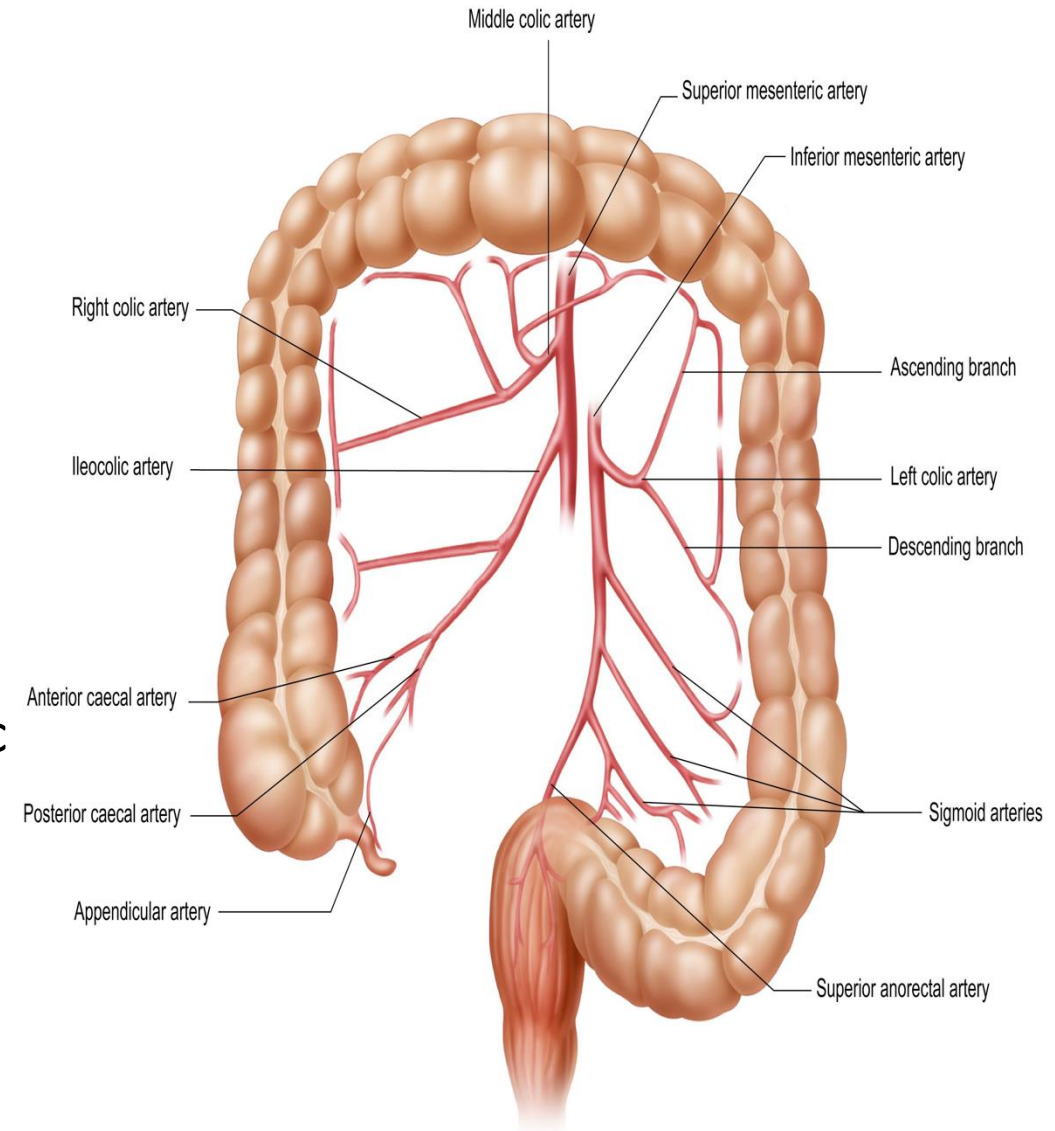


TRANSVERSE COLON

- ❑ The transverse colon is the longest (18 inches/45 cm) and most mobile part of the large intestine,
- ❑ Its intraperitoneal part has a mesentery known as the transverse mesocolon.
- ❑ **Blood Supply:** The middle colic artery, a branch of the superior mesenteric artery, and The left colic artery, a branch of the inferior mesenteric artery
- ❑ **The venous drainage** corresponds to the arterial supply and drains into the superior and inferior mesenteric veins.
- ❑ **Lymph Drainage:** The proximal two-thirds of the transverse colon drain into the colic nodes and subsequently into the superior mesenteric nodes. The distal third drains into the colic nodes and then into the inferior mesenteric nodes.

DESCENDING COLON

- ❑ The descending colon measures approximately 25 cm in length, It extends from the left colic flexure to the pelvic brim, where it transitions into the pelvic colon (sigmoid)
- ❑ It is located in the left paracolic gutter
- ❑ **Blood Supply** : The left colic and sigmoid branches of the inferior mesenteric artery supply this area
- ❑ The corresponding **veins drain** into the inferior mesenteric vein
- ❑ **Lymphatic drainage** occurs through the colic lymph nodes and the inferior mesenteric nodes



THE SIGMOID COLON

- ❑ The sigmoid colon is approximately 15 inches (37.5 cm) long and connects the descending colon with the rectum . S-shaped (from the Greek letter Sigma, representing the S-shape)
- ❑ It extends from the lower end of the descending colon at the left pelvic inlet to the pelvic surface of the third sacral vertebra, where it becomes continuous with the rectum.
- ❑ The sigmoid branches of the inferior mesenteric artery supply blood to the sigmoid colon
- ❑ The accompanying veins drain into the inferior mesenteric vein, which subsequently joins the portal venous system
- ❑ Lymph Drainage: Lymphatic drainage occurs through nodes along the course of the sigmoid arteries, from these nodes, lymph travels to the inferior mesenteric nodes.

CLINICAL ANATOMY

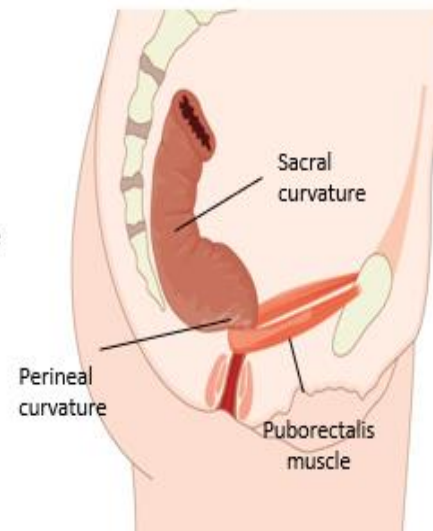
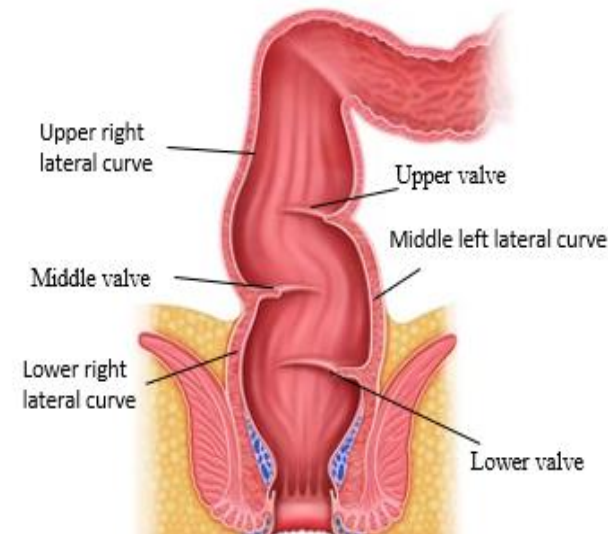
- ❑ The sigmoid colon is often selected as a site for performing a colostomy in patients with carcinoma of the rectum. Its mobility allows the surgeon to bring out a loop of colon
- ❑ Volvulus Because of its extreme mobility, the sigmoid colon sometimes rotates around its mesentery, is referred to as volvulus.
- Diverticula are common along the sigmoid colon. In conditions such as diverticulitis or ulcerative colitis, it may adhere to nearby organs and form internal fistulas.
- Because the sigmoid colon is close to the anus, its mucosa can be examined directly (e.g., by endoscopy) for abnormalities.

RECTUM

- ❑ The rectum is a wide tube, approximately 13 cm long and 4 cm in diameter
- ❑ It begins in front of the third sacral vertebra and ends at the tip of the coccyx.
- ❑ The rectum is located in the posterior part of the lesser pelvis, positioned behind the urinary bladder in males—separated by the rectovesical pouch—and behind the uterus in females, separated by the uterovesical pouch (Douglas' pouch) They act as a barrier to limit the spread of infections.

Although the rectum is a part of the large intestine, it is devoid of taenia coli, sacculations, and appendices epiploicae

The rectum has five curvatures: three laterally and two Antero-posteriorly

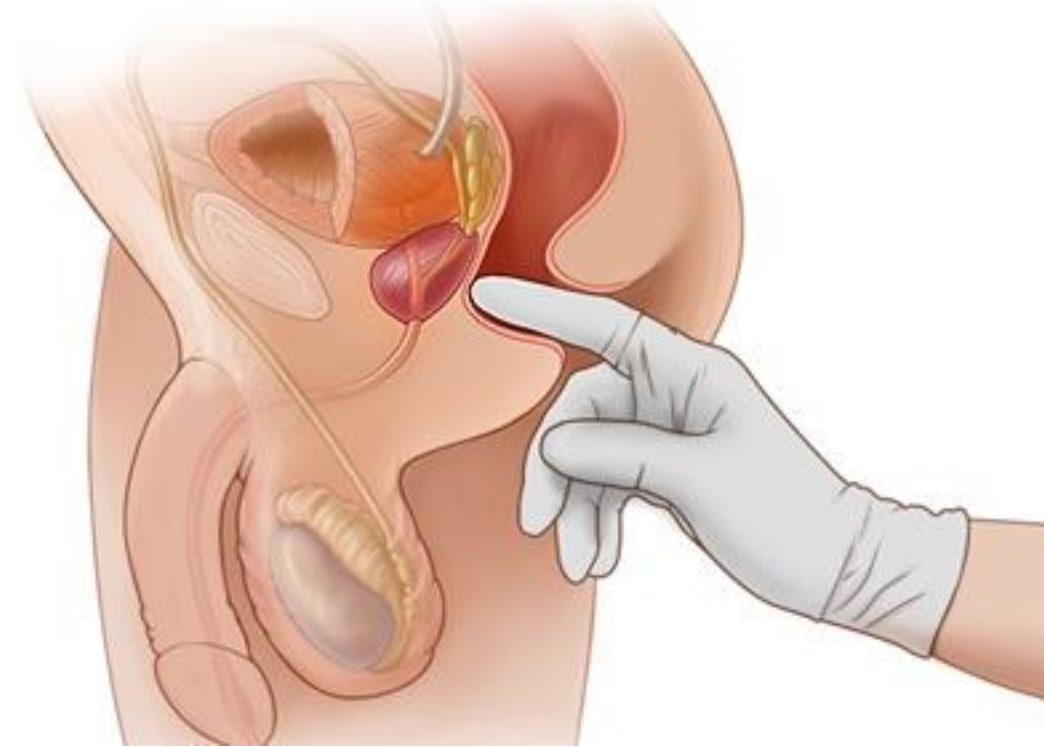


- ❑ In its lower part, the rectum forms a dilation known as the rectal ampulla

Relations

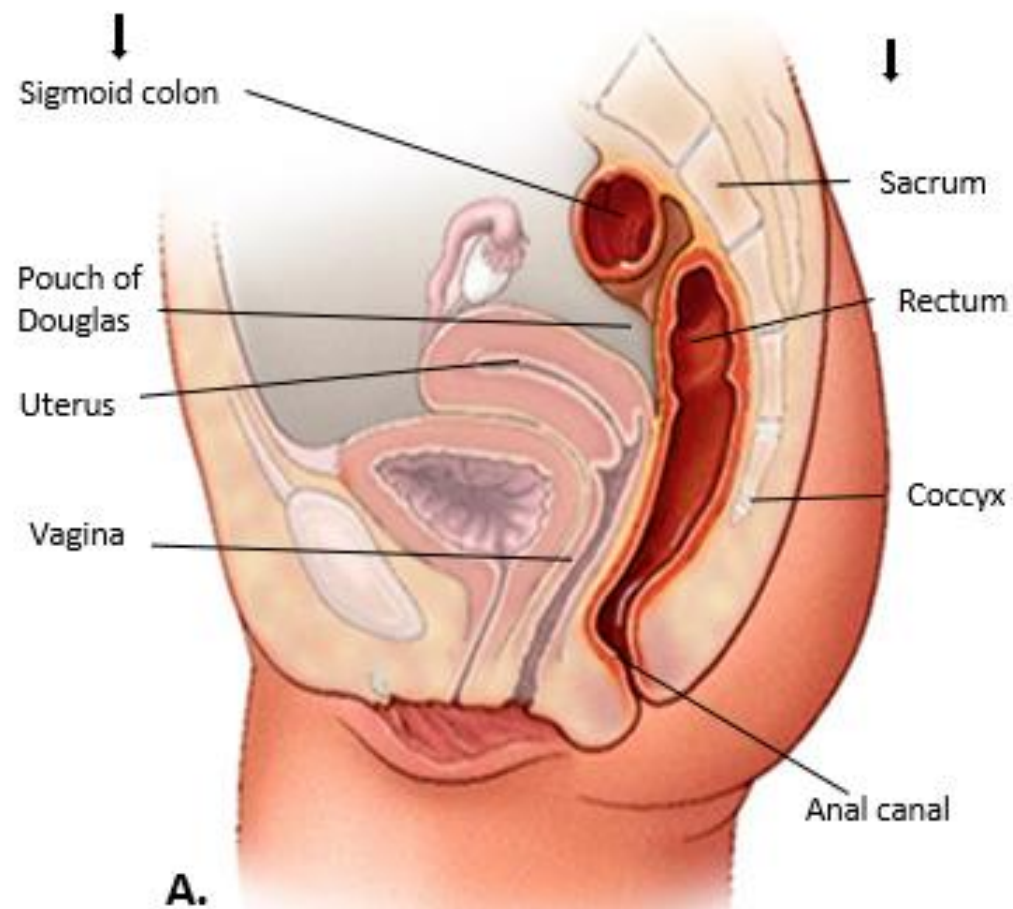
The relations of the rectum are clinically important because a significant amount of information can be gathered about the organs surrounding the rectum through A digital rectal examination.

A digital rectal examination (DRE) is conducted by inserting a gloved, lubricated finger into the rectum



❑ This allows for palpation of potential abnormalities, such as lumps, tumors, enlargements, tissue hardening, hemorrhoids, rectal carcinoma, prostate cancer, seminal vesicles, the ampulla of the ductus deferens, bladder, uterus, cervix, ovaries, anorectal abscesses, polyps, chronic constipation, and other irregularities.

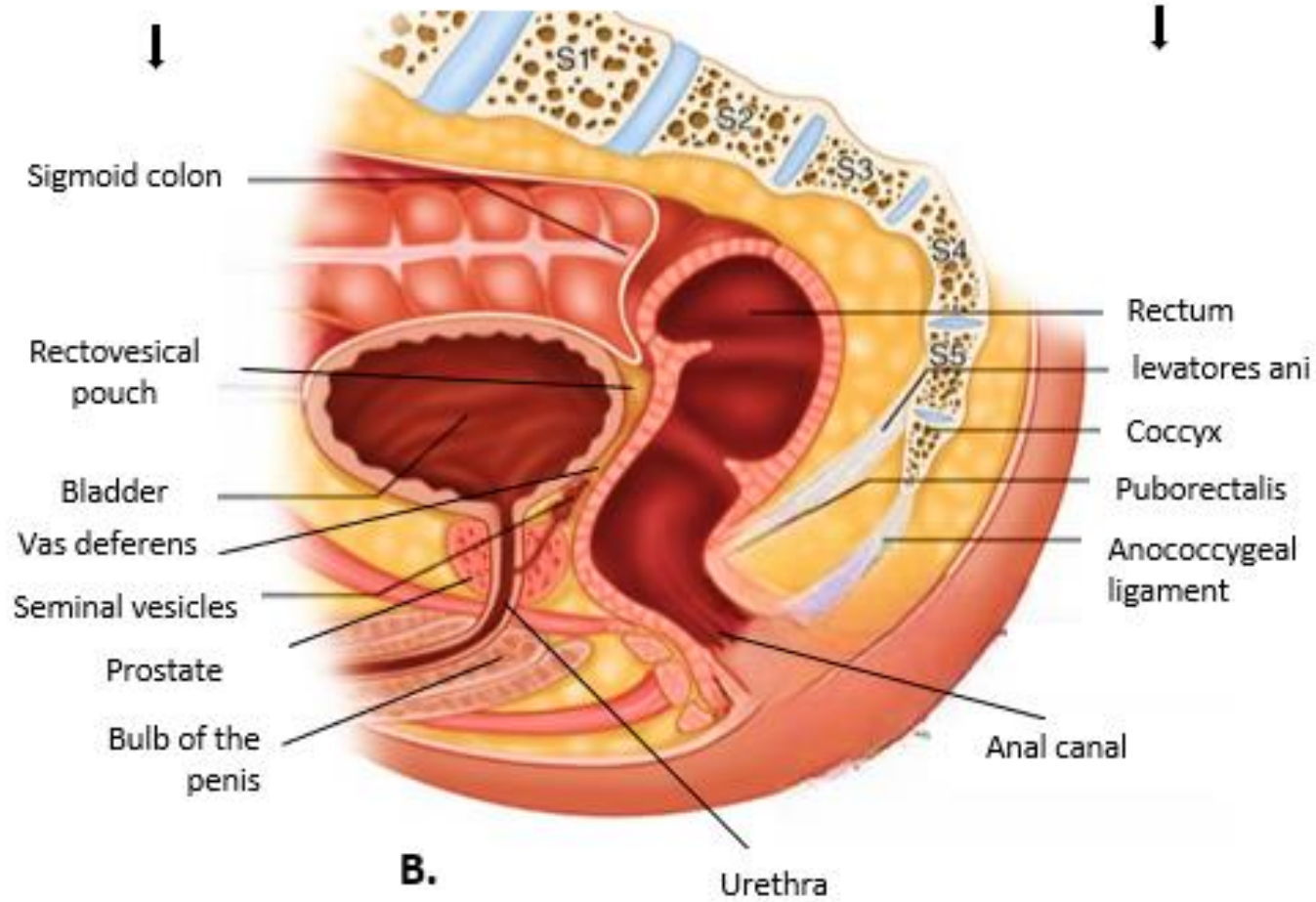
Anterior relations



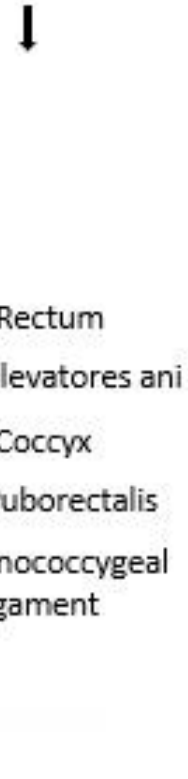
Posterior relations



Anterior relations

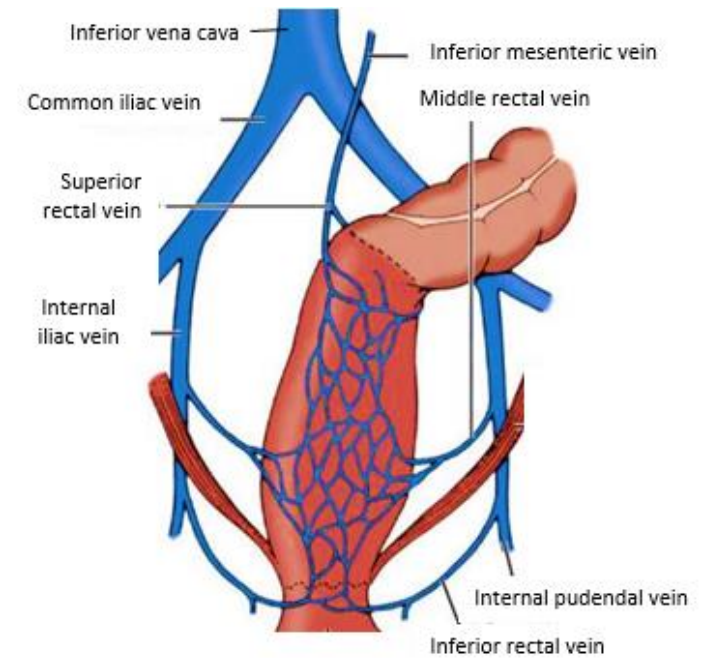
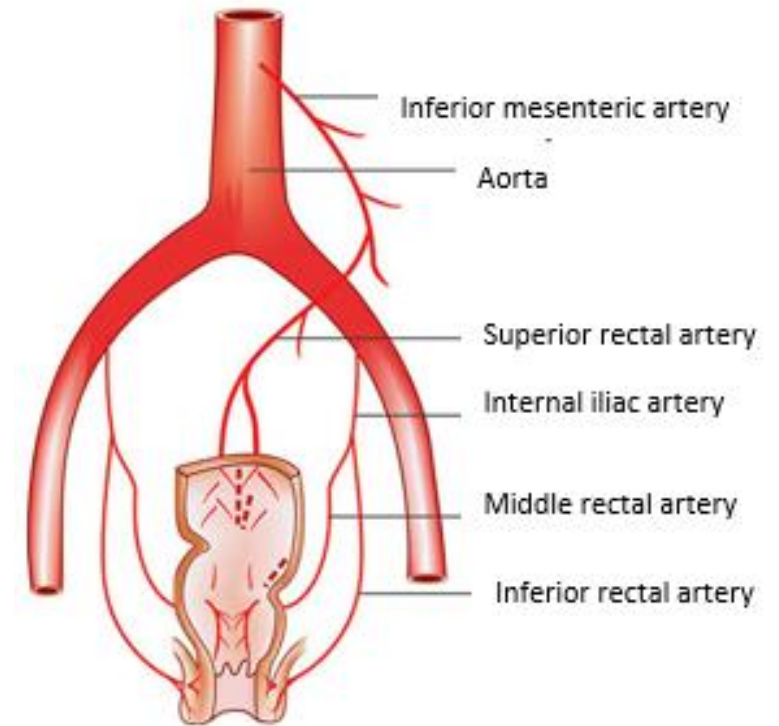


Posterior relations



Blood Supply

- **Arteries:** The superior, middle, and inferior rectal arteries, supply the rectum.
- **Veins:** The veins of the rectum correspond to the arteries. The superior rectal vein is a tributary of the portal circulation, draining into the inferior mesenteric vein. The middle and inferior rectal veins drain into the internal iliac and internal pudendal veins,

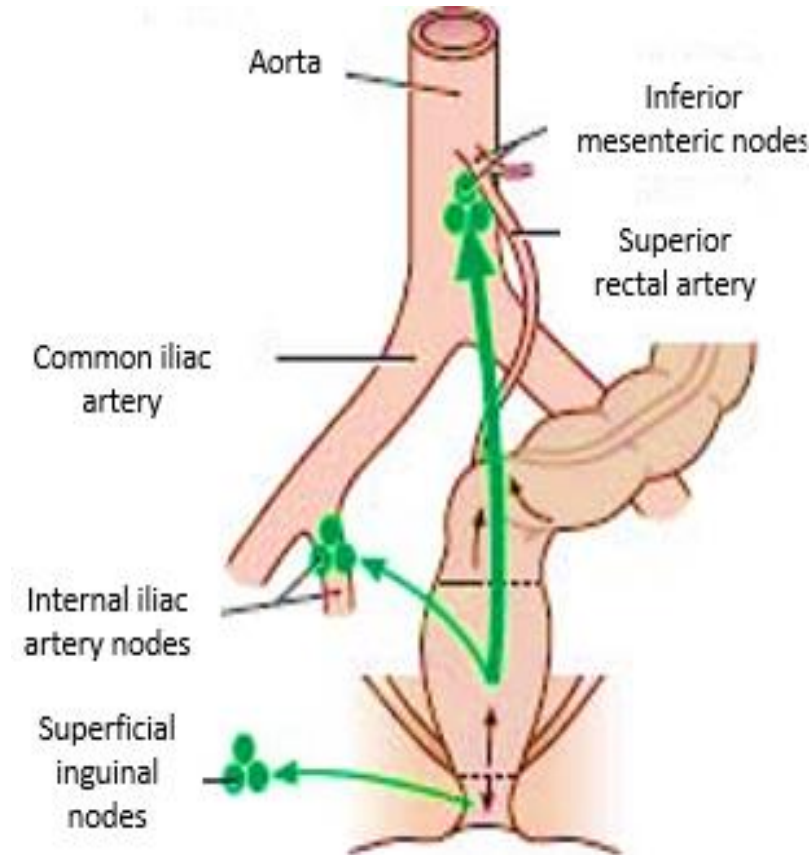


□ Lymphatic drainage

- The lymph vessels of the upper rectum drain first into the pararectal nodes and then into the inferior mesenteric nodes.
- Lymph vessels from the lower part of the rectum follow the middle rectal artery to the internal iliac nodes

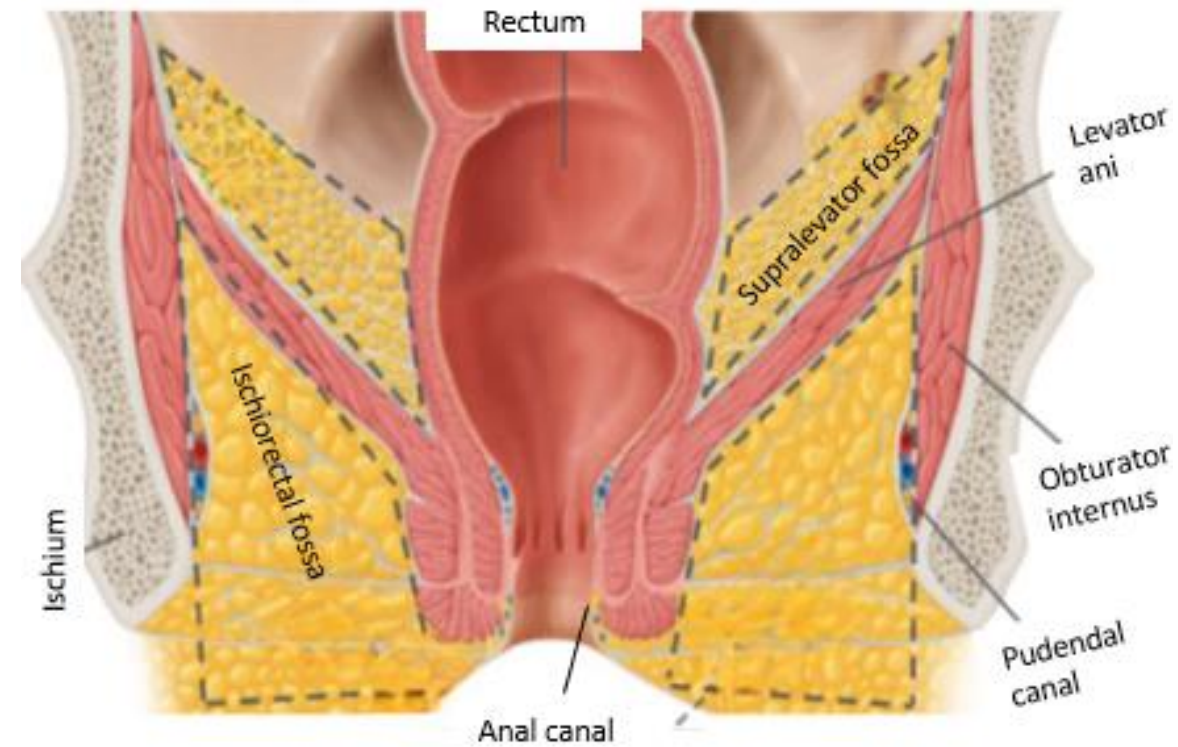
Rectal Cancer:

Rectal cancer may spread via lymphatic vessels. The superior rectal is a tributary of the portal vein, meaning that rectal cancer can metastasize to the liver. The cancer may also penetrate the rectal wall posteriorly, invading the sacral plexus and causing sciatica, or invade laterally into the ureter and anteriorly into the vagina, uterus, bladder, prostate, or seminal vesicles.



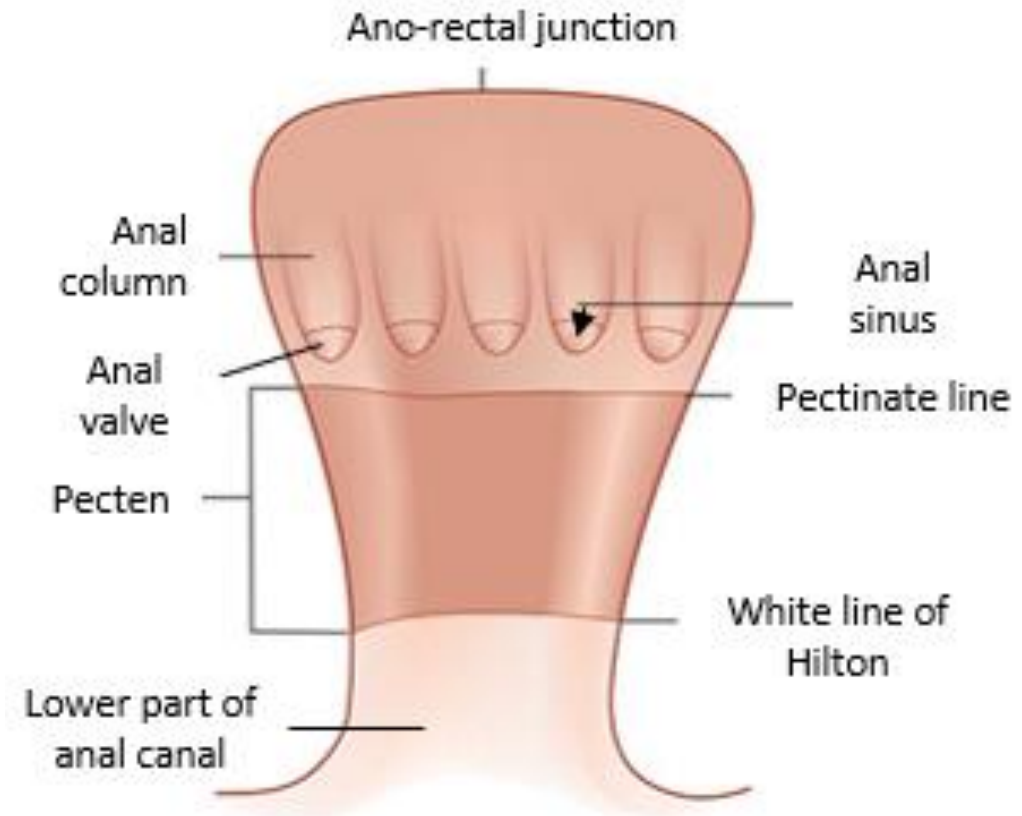
Anal canal

- ❑ The anal canal is the terminal part of the large intestine; it 4 cm in length and is located in the perineum and ending at the anus.
- ❑ It is encircled by an inner involuntary sphincter and an outer voluntary sphincter, which maintain closure of the anal canal except during defecation.
- ❑ The anal canal is situated within the anal triangle of the perineum, positioned between the right and left fat-filled ischiorectal fossae
- ❑ These fossae accommodate the expansion of the anal canal during the process of defecation.



Interior of the anal canal

- ❑ Anatomically, the anal canal is divided into an upper two-thirds (visceral portion), which is part of the intestine, and a lower one-third (somatic portion), which is part of the perineum, by the pectinate line.
- ❑ The upper and lower sections of the anal canal differ in terms of embryological development, blood supply, lymphatic drainage, and nerve supply. Table outlines the differences between the upper and lower anal canals.



Feature	Upper Anal Canal	Lower Anal Canal
Development	Derived from the endoderm.	Derived from the ectoderm.
Innervation	Autonomic (sympathetic and parasympathetic) innervation via the inferior hypogastric plexus.	Somatic innervation via the inferior rectal nerve (branch of the pudendal nerve).
Arterial Supply	Superior rectal artery (branch of the inferior mesenteric artery).	Inferior rectal artery (branch of the internal pudendal artery).
Epithelial Lining	Simple columnar epithelium (mucosal lining).	Stratified squamous epithelium (non-keratinized to keratinized closer to the anal verge).
Venous Drainage	Drains into the portal venous system via the superior rectal vein.	Drains into the systemic venous system via the inferior rectal vein.
Lymphatic Drainage	Drains to the internal iliac lymph nodes.	Drains to the superficial inguinal lymph nodes.
Hemorrhoids	Internal hemorrhoids (painless due to autonomic innervation).	External hemorrhoids (painful due to somatic innervation).

Anal sphincters

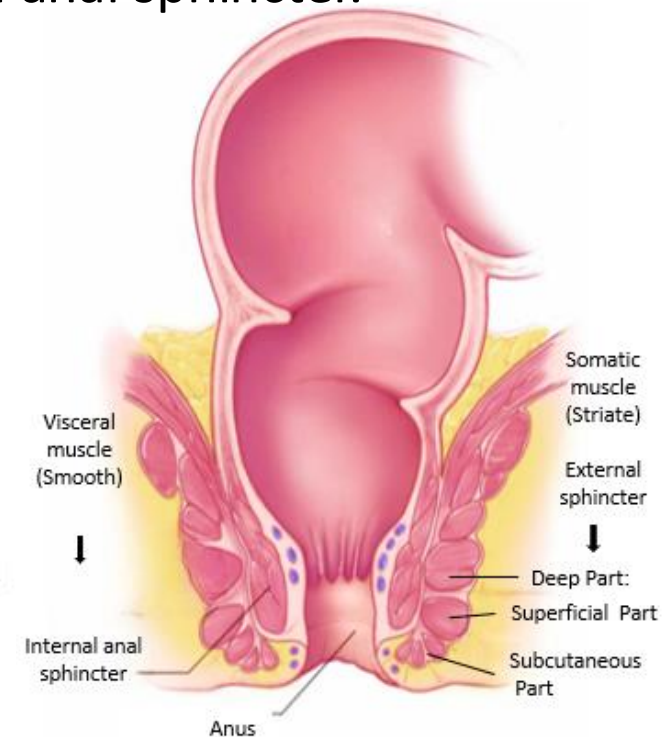
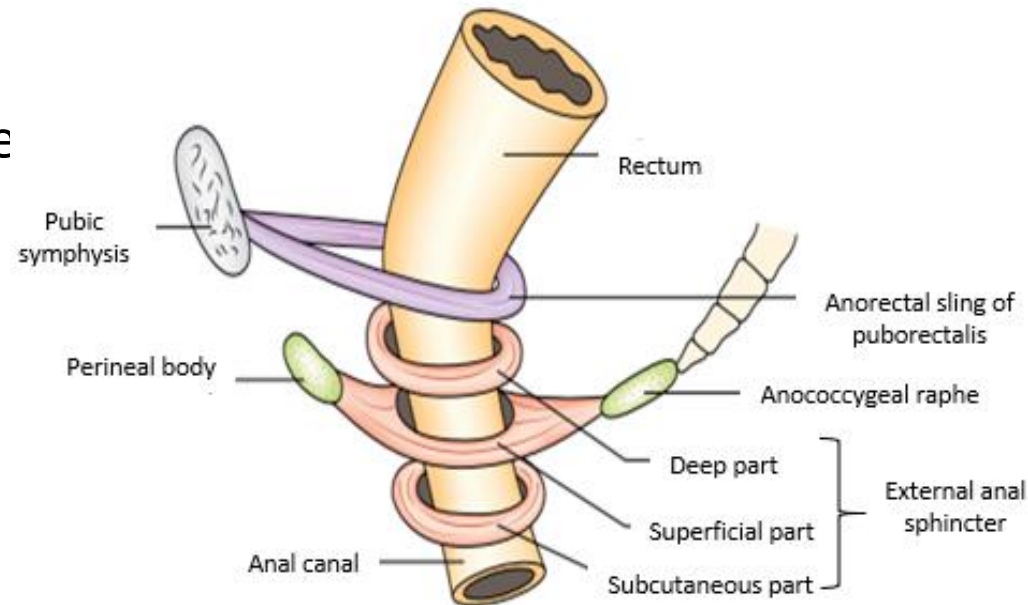
The anal canal has two types of sphincters (internal and external), which provide a powerful sphincteric mechanism at the distal end of the gastrointestinal tract.

Internal sphincter:

It is formed by a thickened circular muscle coat of the anal canal and surrounds the upper two-thirds of the canal and surrounded by the deep and superficial parts of the external anal sphincter.

External sphincter

The external anal sphincter, as the name implies It surrounds the whole length of the anal canal and consists of three parts: deep, superficial, and subcutaneous.



Blood Supply

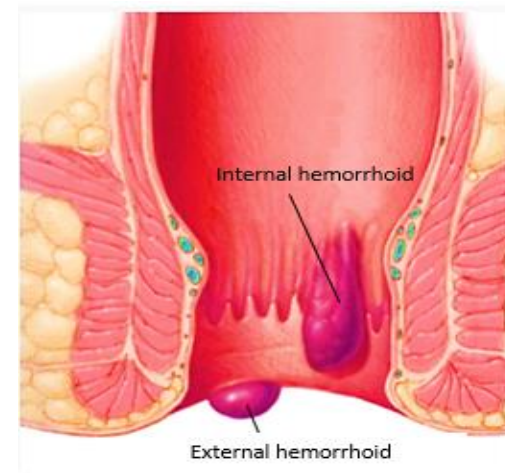
- ❑ **Arteries:** The upper part of the anal canal is supplied by the superior rectal artery, while the lower part is supplied by the inferior rectal artery.
- ❑ **Veins:** The upper part of the anal canal drains into the portal venous system via the superior rectal vein, and the lower part drains into the caval (systemic) system through the inferior rectal veins.
- ❑ **Lymphatic Drainage:** The upper part of the anal canal drains into the internal iliac nodes, whereas the lower part drains into the horizontal group of superficial inguinal nodes.

Clinical correlation HEMORRHOIDS (OR PILES)

Hemorrhoids are dilatations of the anal submucosal and perianal venous plexuses.

They are classified into two types: internal and external

- **Internal hemorrhoids (or true piles):** are saccular dilatations of the tributaries of the superior rectal (hemorrhoidal) vein, located above the pectinate line in cases of portal obstruction and occur in the upper half, they are painless and sensitive only to stretch. This may explain why large internal hemorrhoids can cause an aching sensation rather than acute pain.
- **External hemorrhoids (or false piles):** are dilatations of the tributaries of the inferior rectal vein located below the pectinate line, are very painful and typically do not bleed during straining in defecation. Innervated by the inferior rectal nerves, they are sensitive to pain, temperature, touch, and pressure, which explains the pain associated with external hemorrhoids.



Thank You